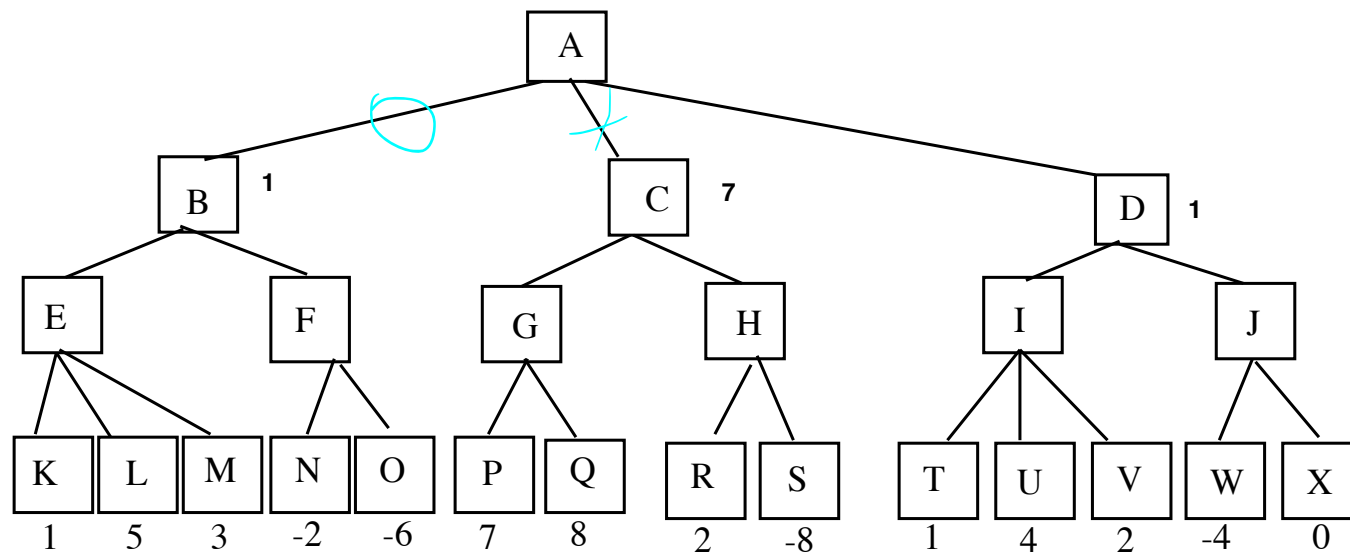


Exercise – min-max and alfa-beta cuts

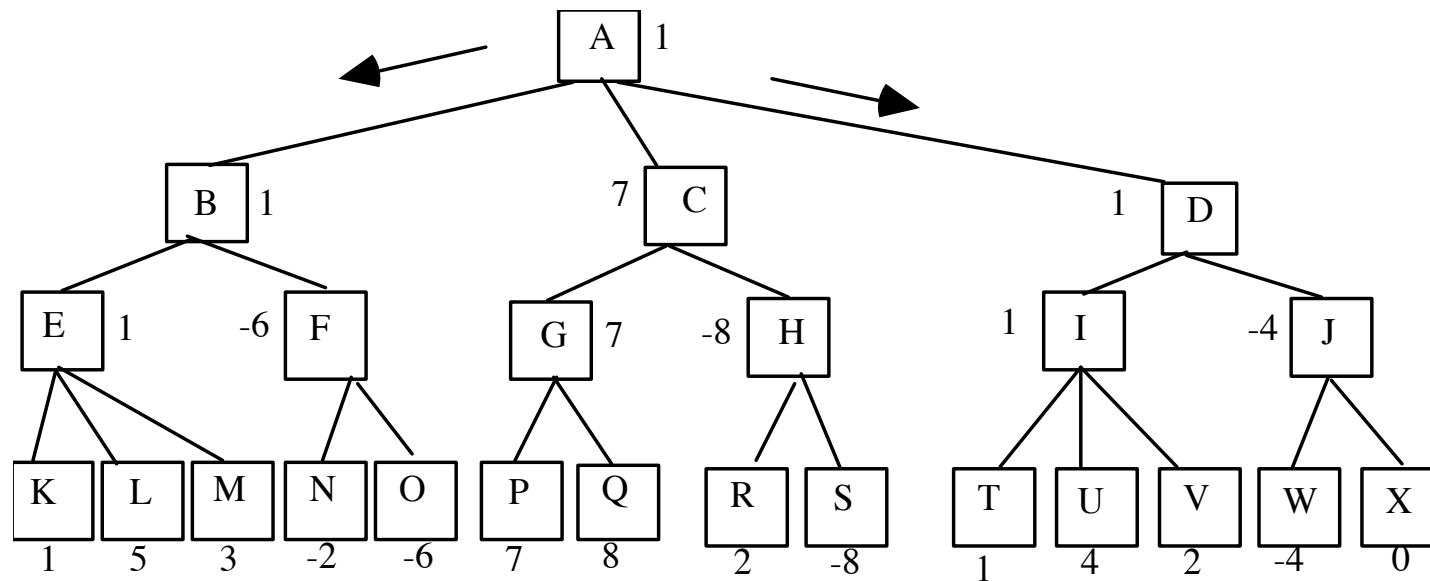
- Consider the following search tree representing a game. The evaluation of the leaves has been given by the first player.



- Suppose that the first player is MIN, which move is the most convenient? First you have to apply the MIN-MAX algorithm. Then alfa beta cuts should be identified

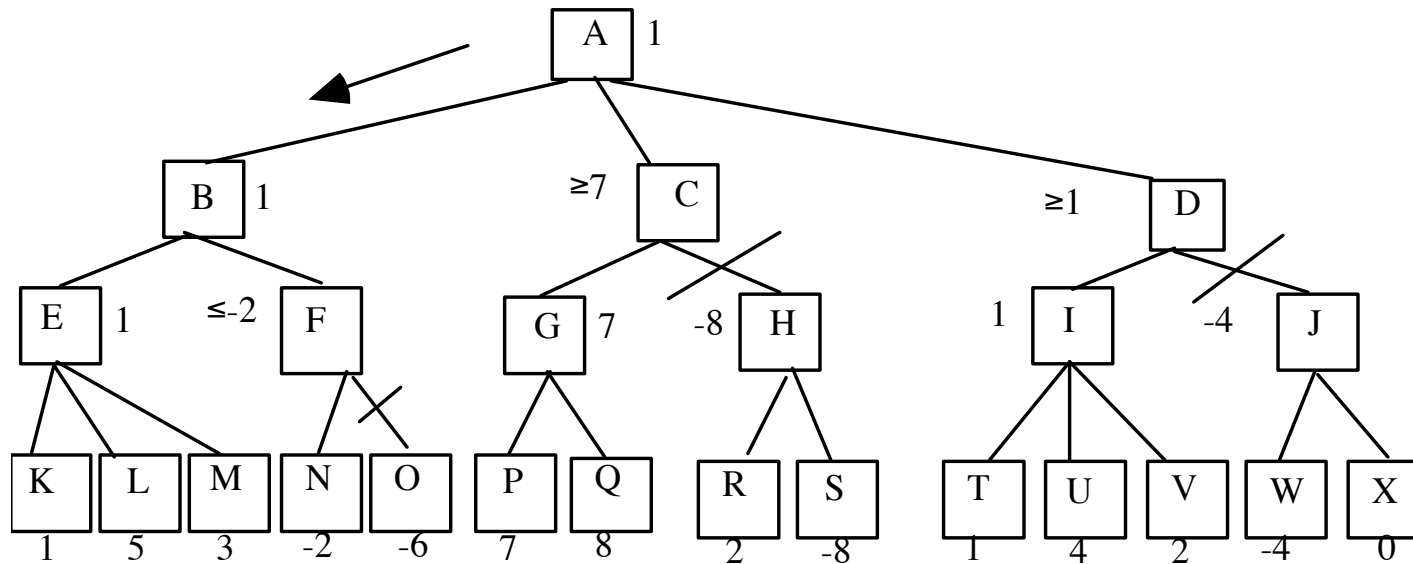
Exercise – min-max and alfa-beta cuts

- Min-Max:



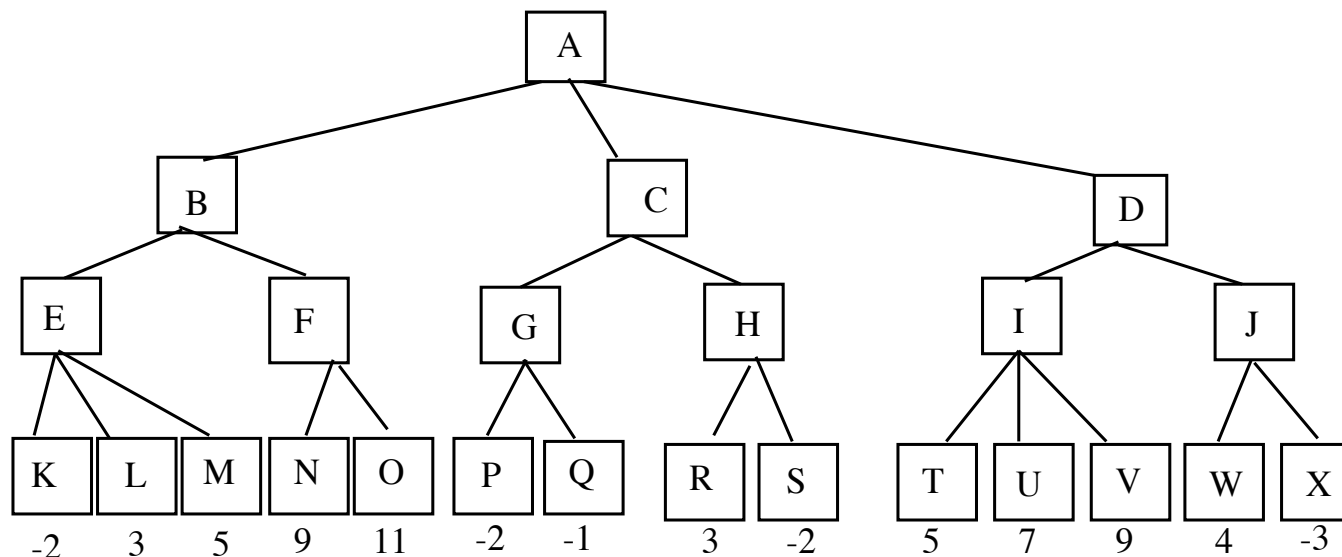
Exercise – min-max and alfa-beta cuts

- Alfa-beta cuts



Exercise – min-max and alfa-beta cuts

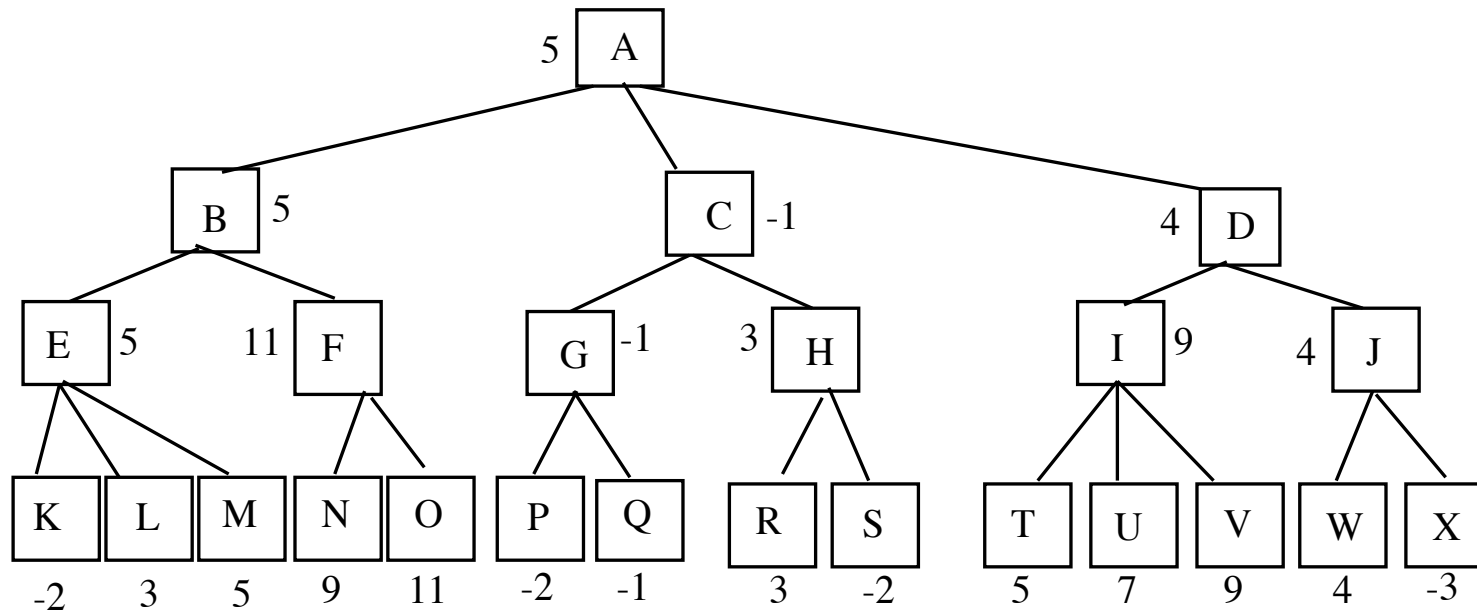
- Consider the following search tree representing a game. The evaluation of the leaves has been given by the first player.



- Suppose that the first player is MAX, which move is the most convenient? First you have to apply the MIN-MAX algorithm. Then alfa beta cuts should be identified

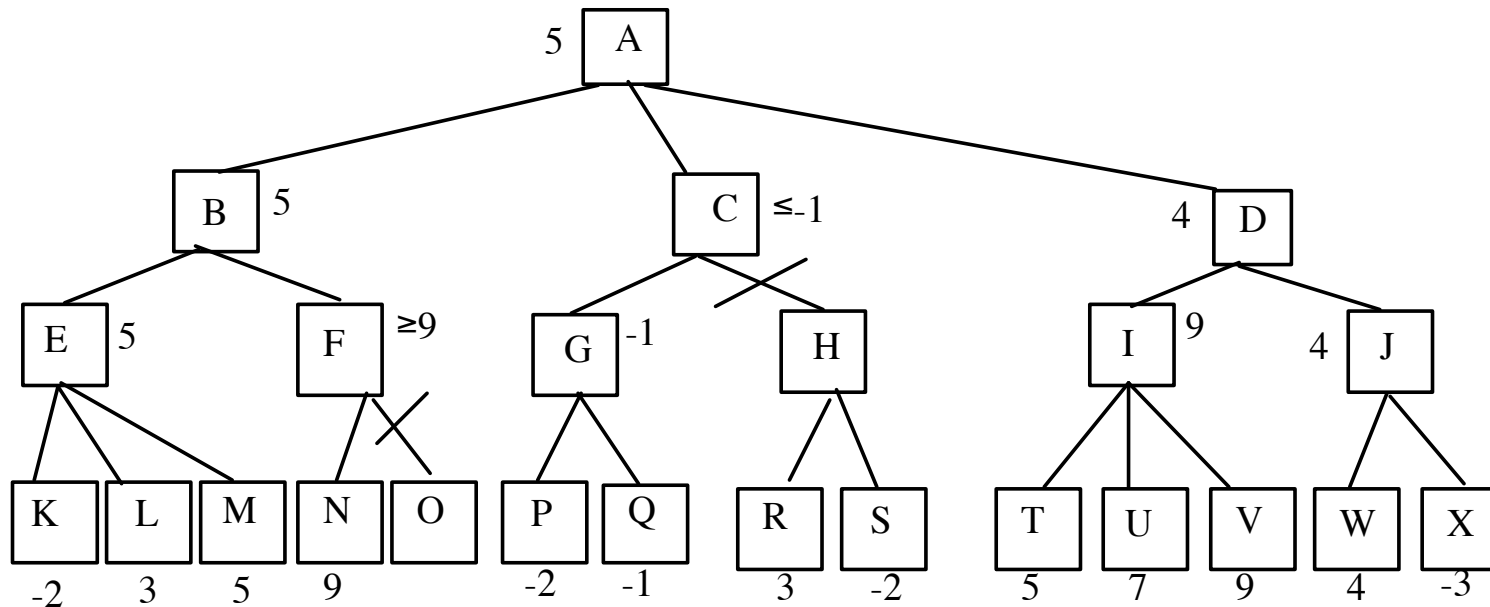
Exercise – min-max and alfa-beta cuts

- Min-Max:

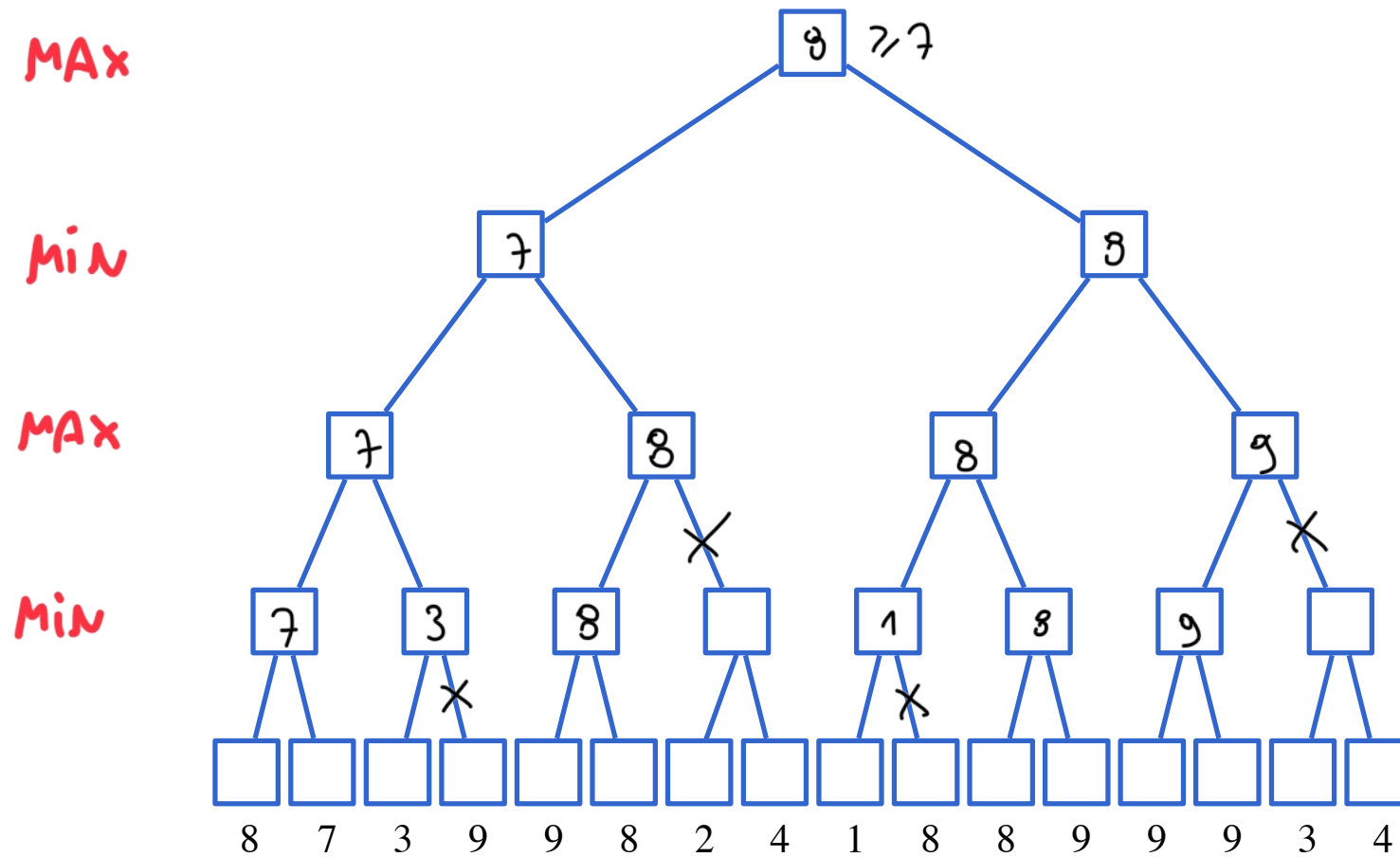


Exercise – min-max and alfa-beta cuts

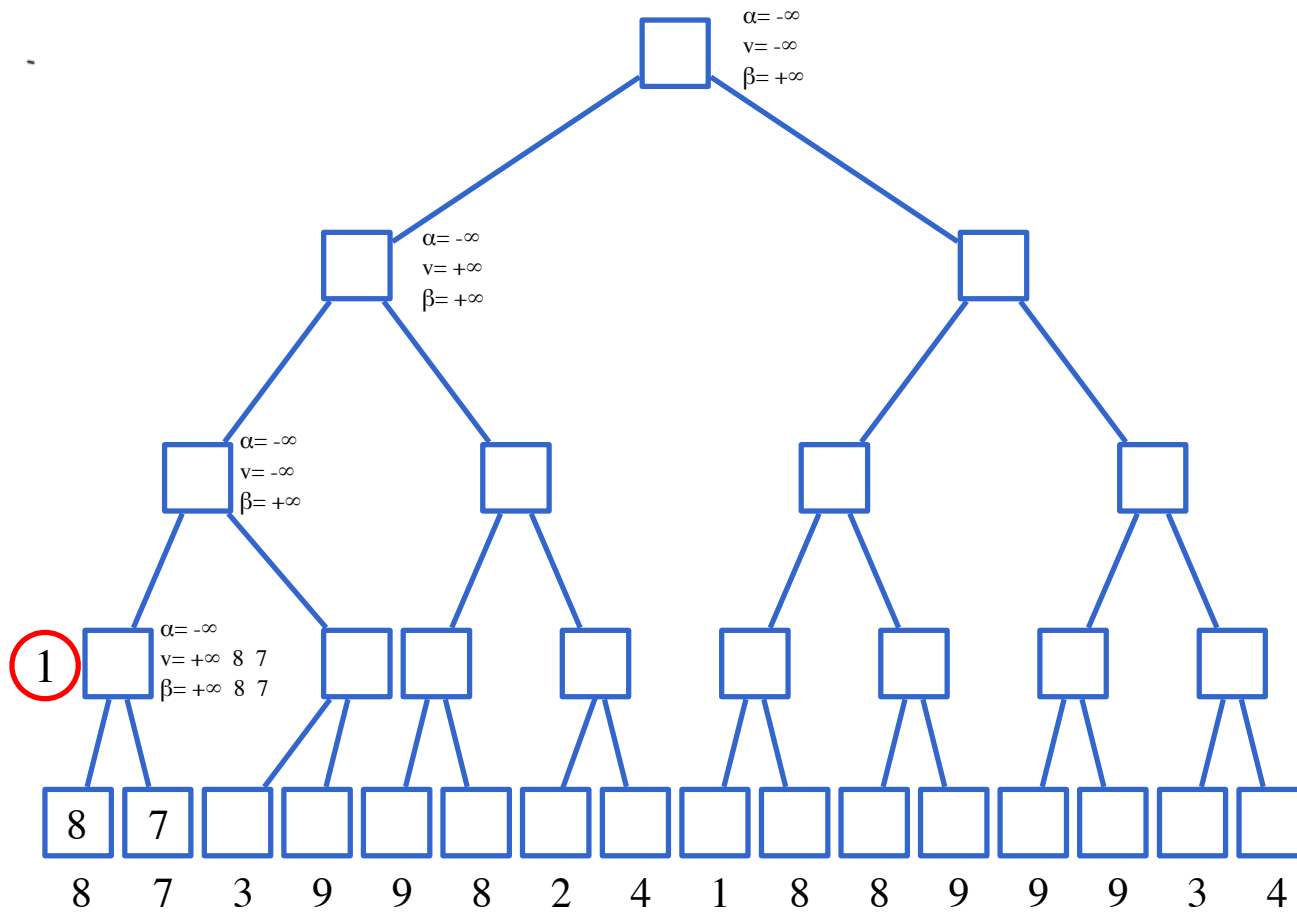
- Alfa-beta cuts:



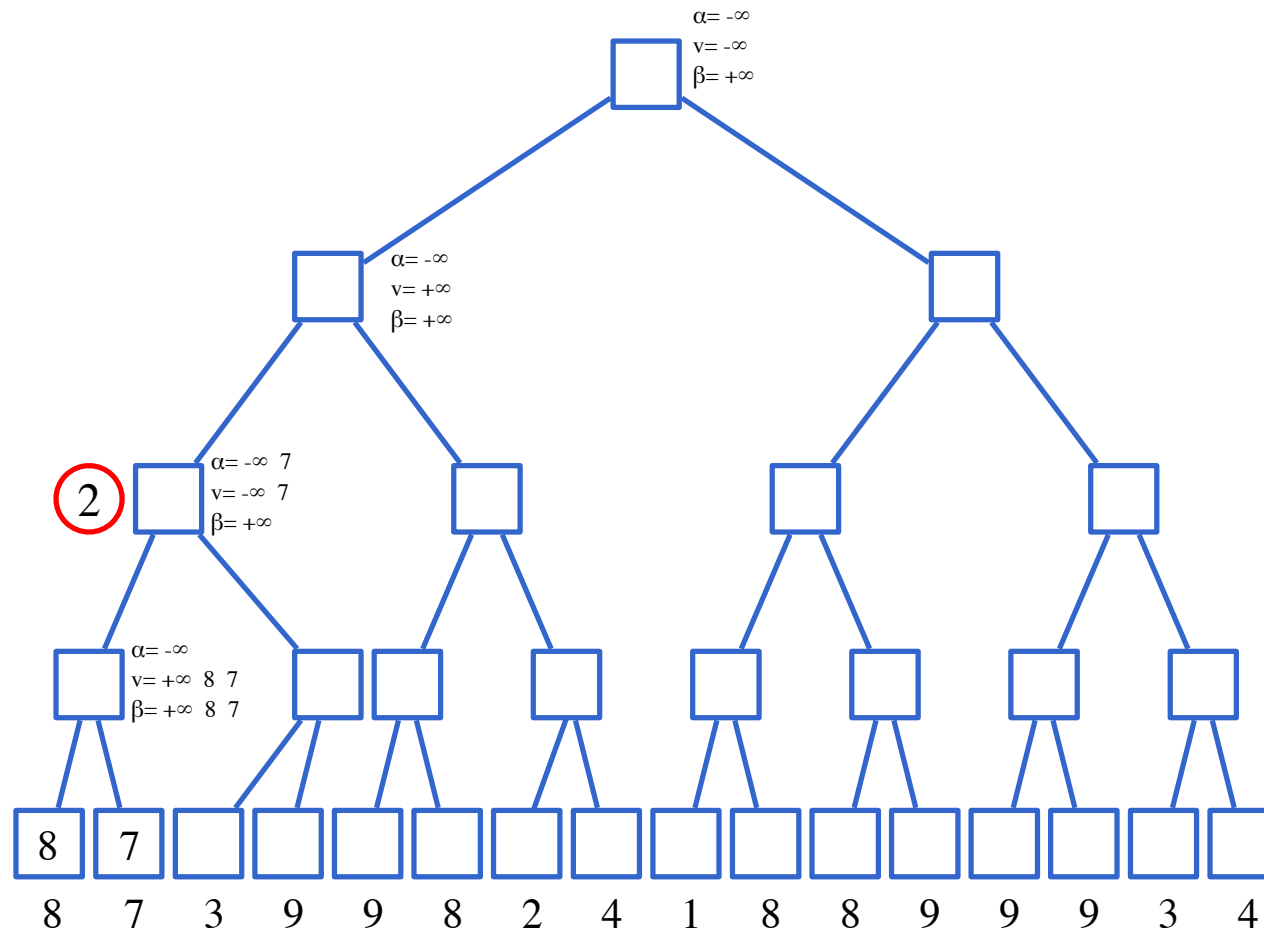
Exam 21/12/2011



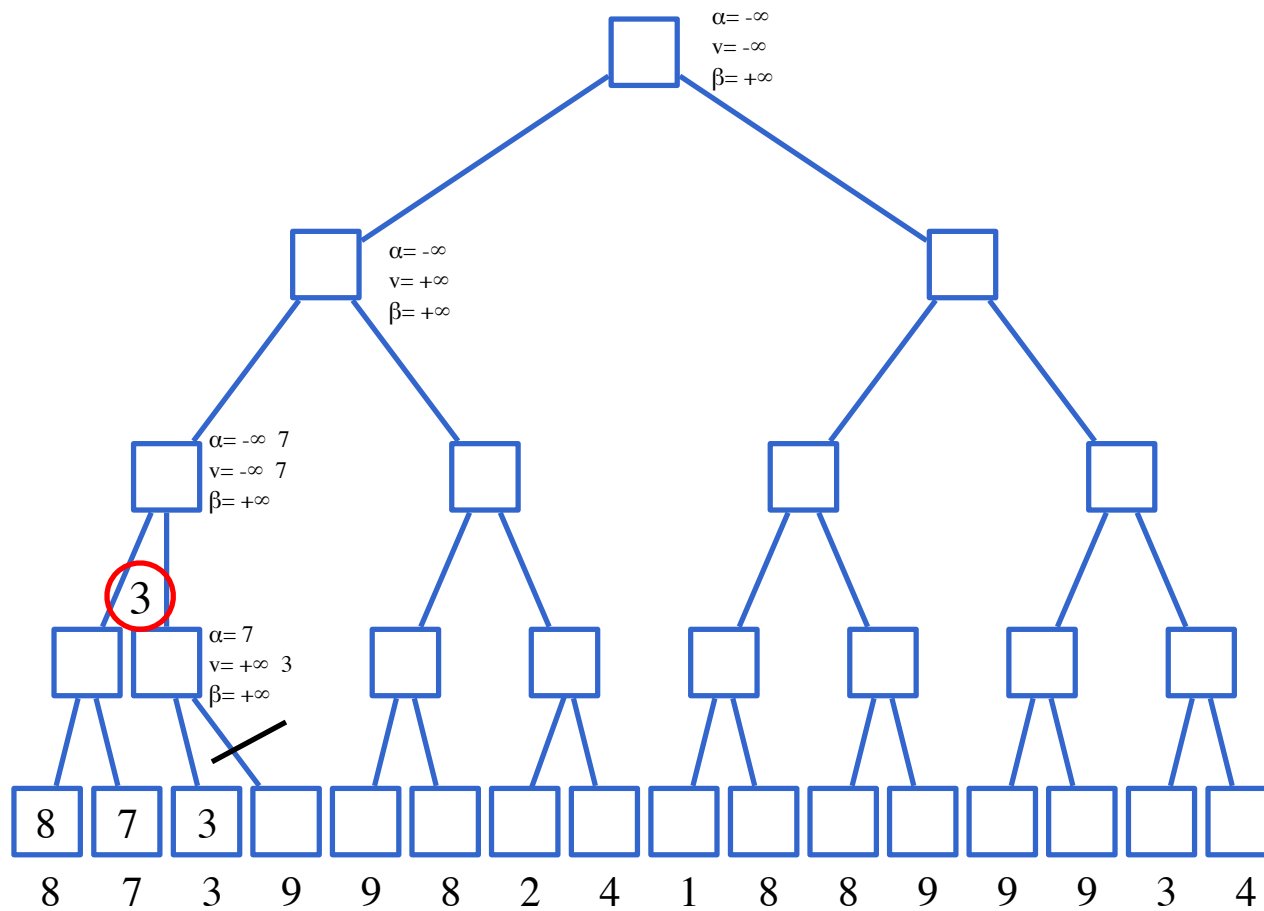
Exam 21/12/2011



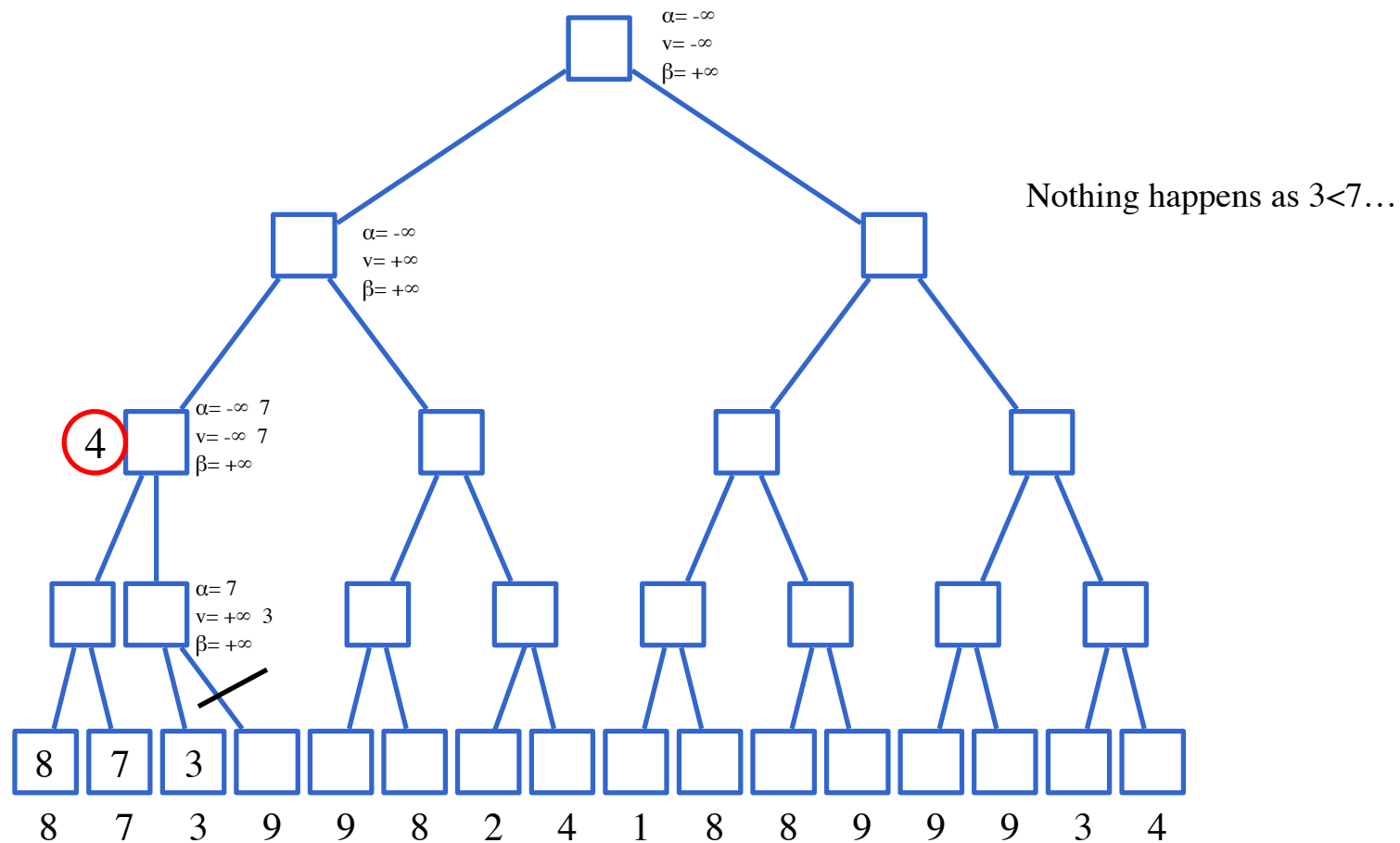
Exam 21/12/2011



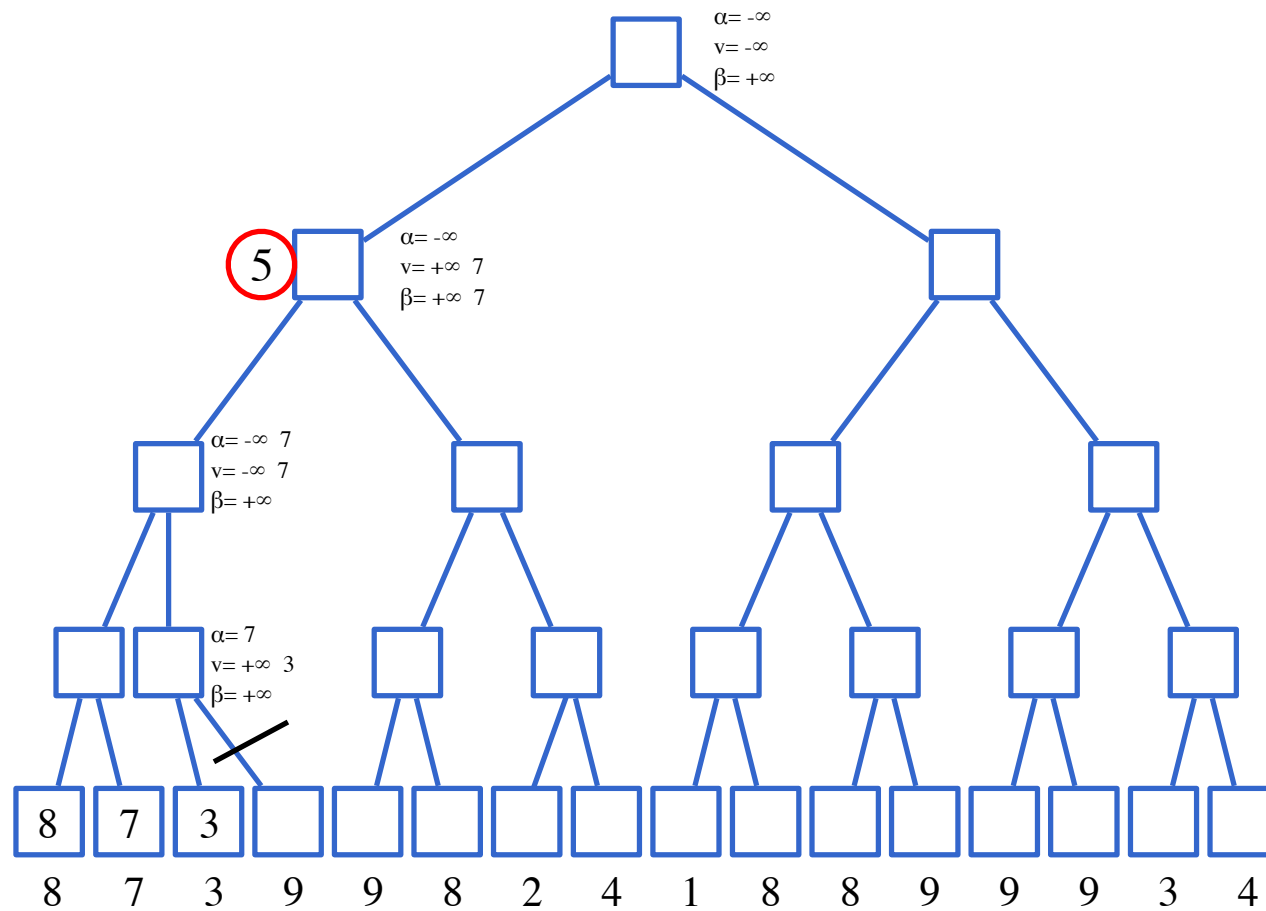
Exam 21/12/2011



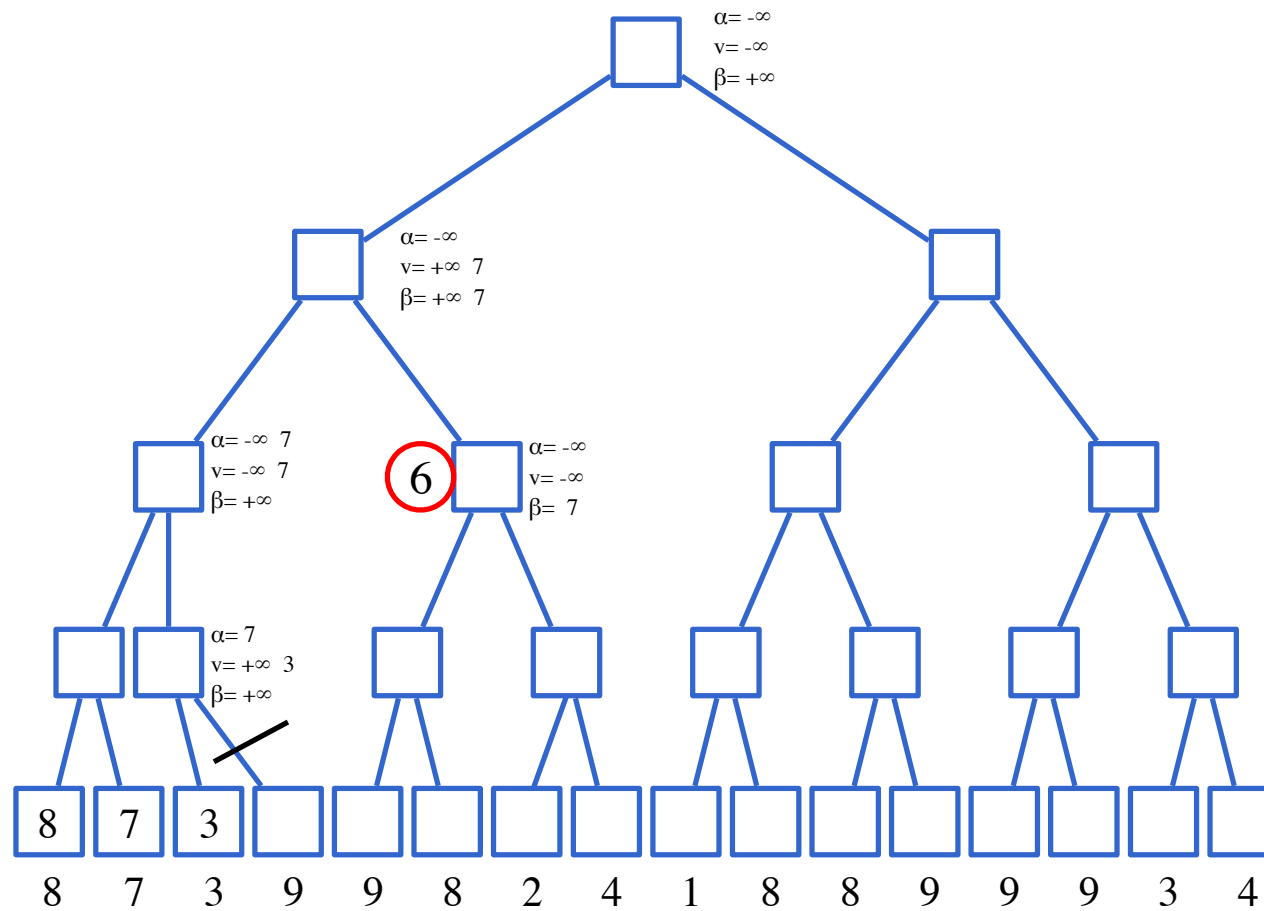
Exam 21/12/2011



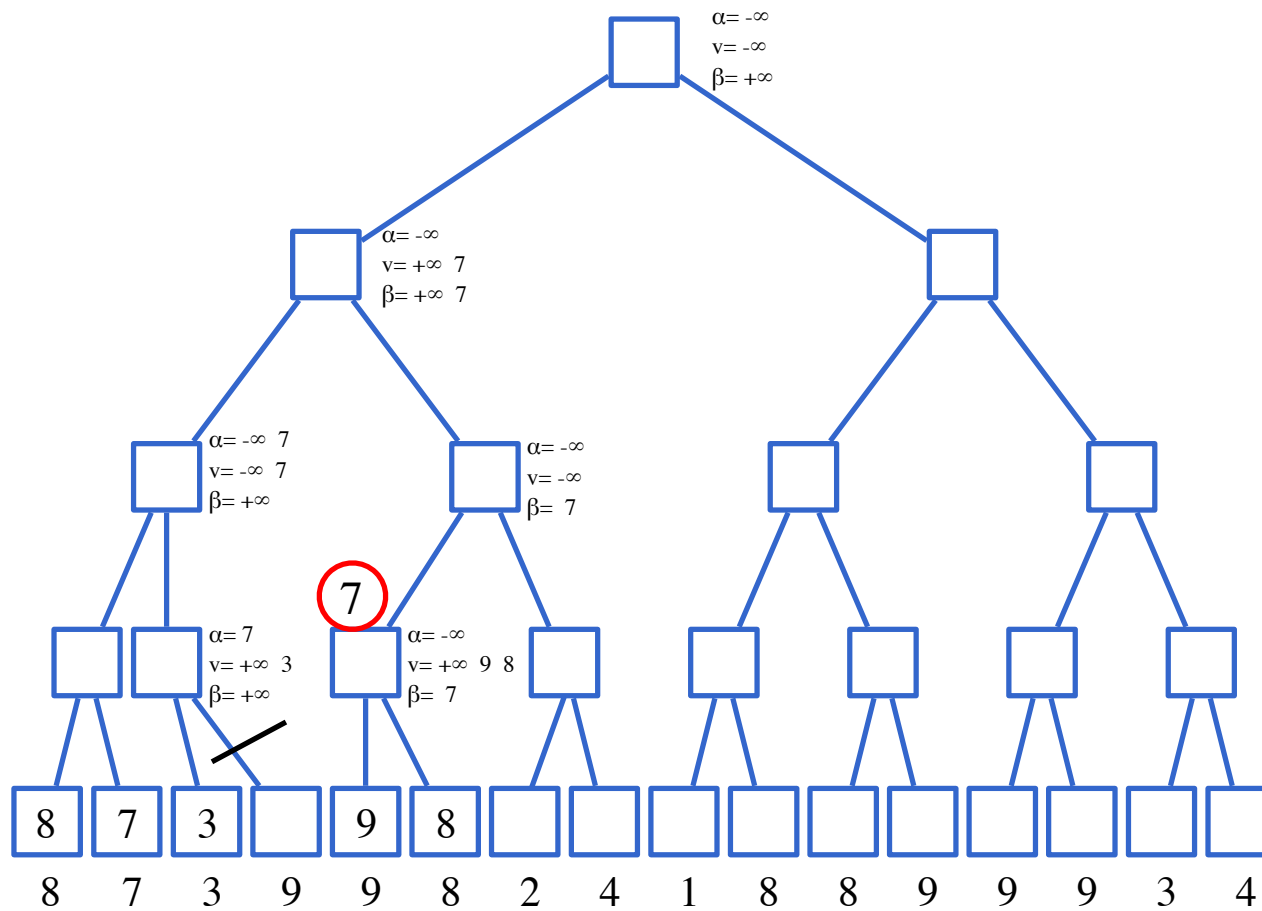
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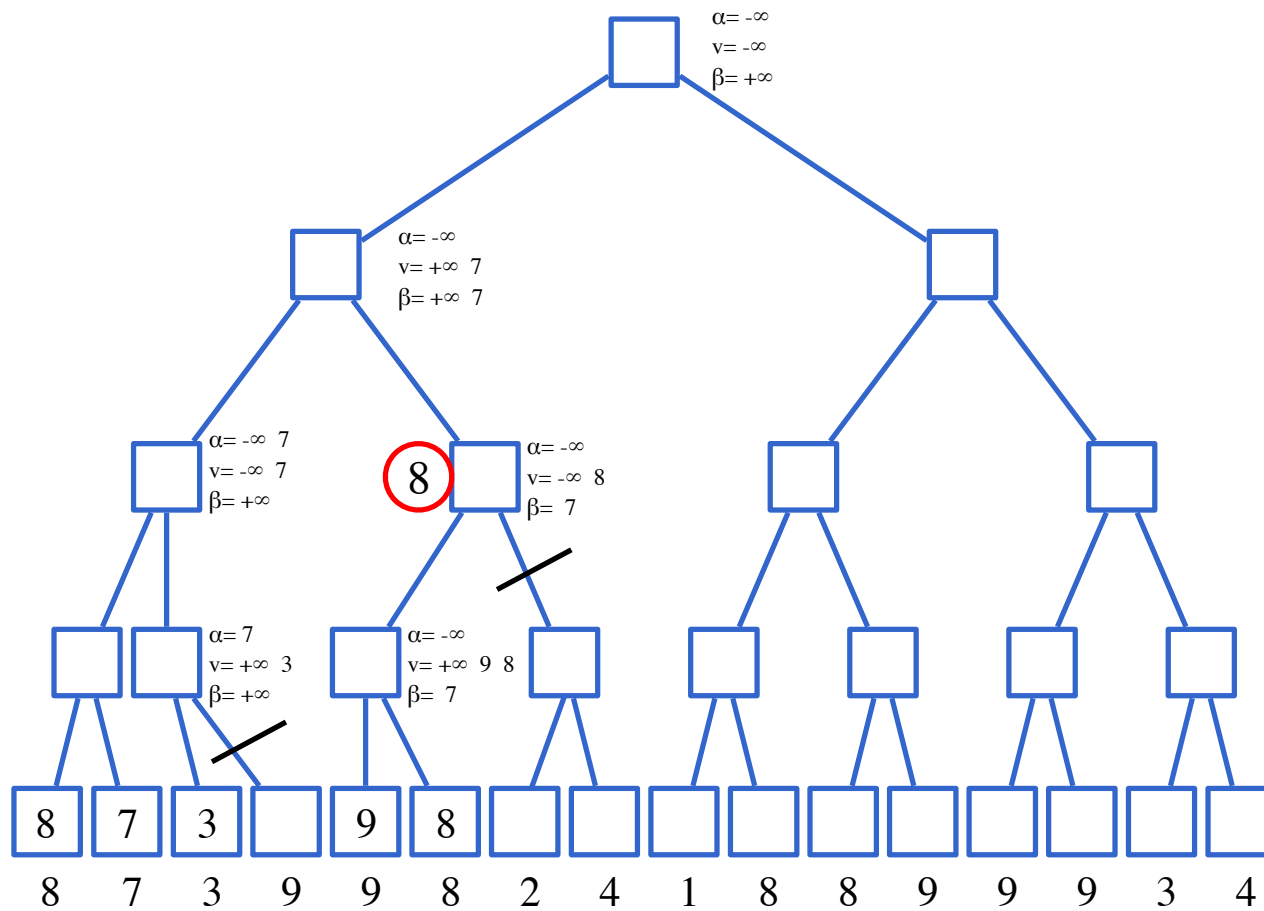
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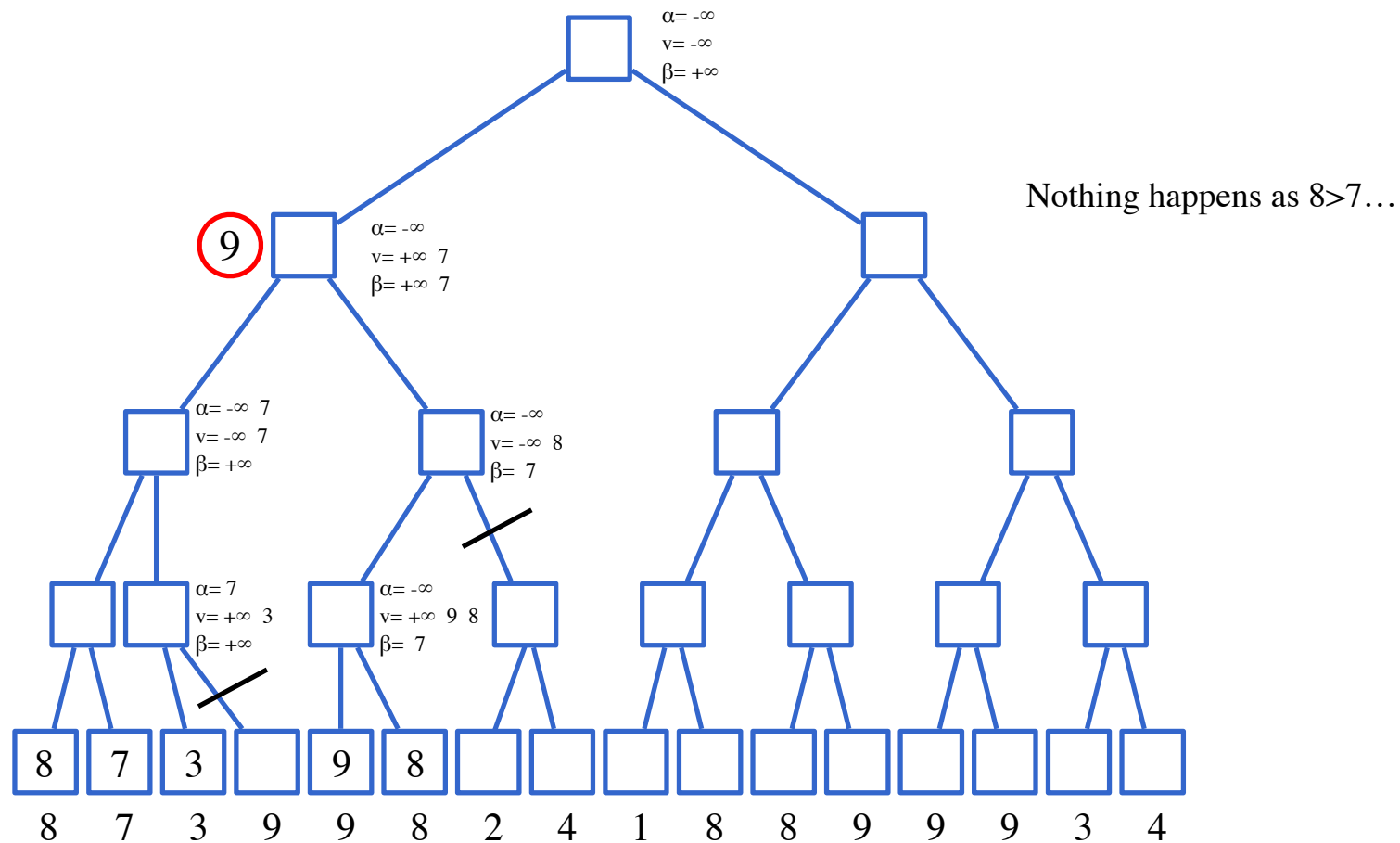
Exam 21/12/2011



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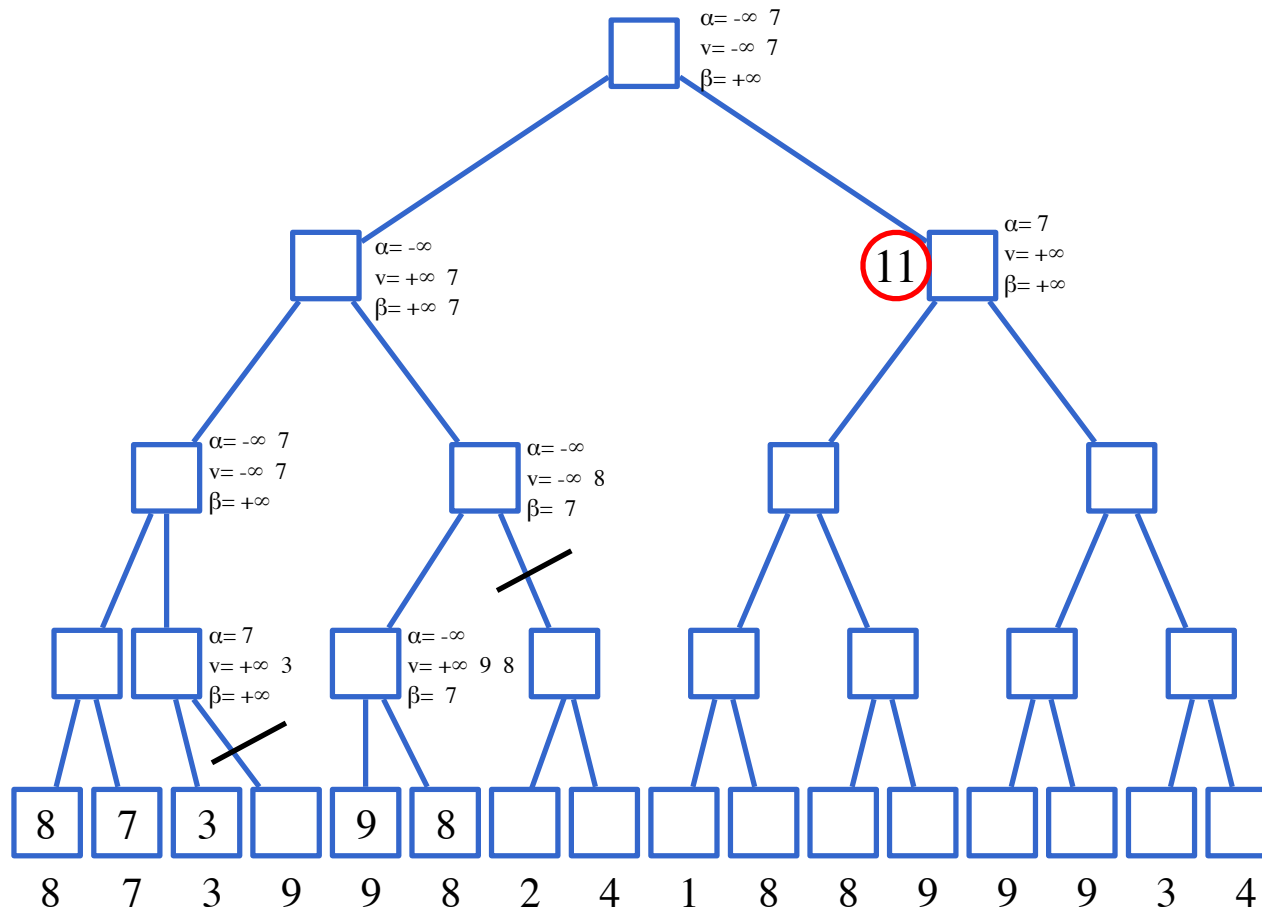


Exam 21/12/2011

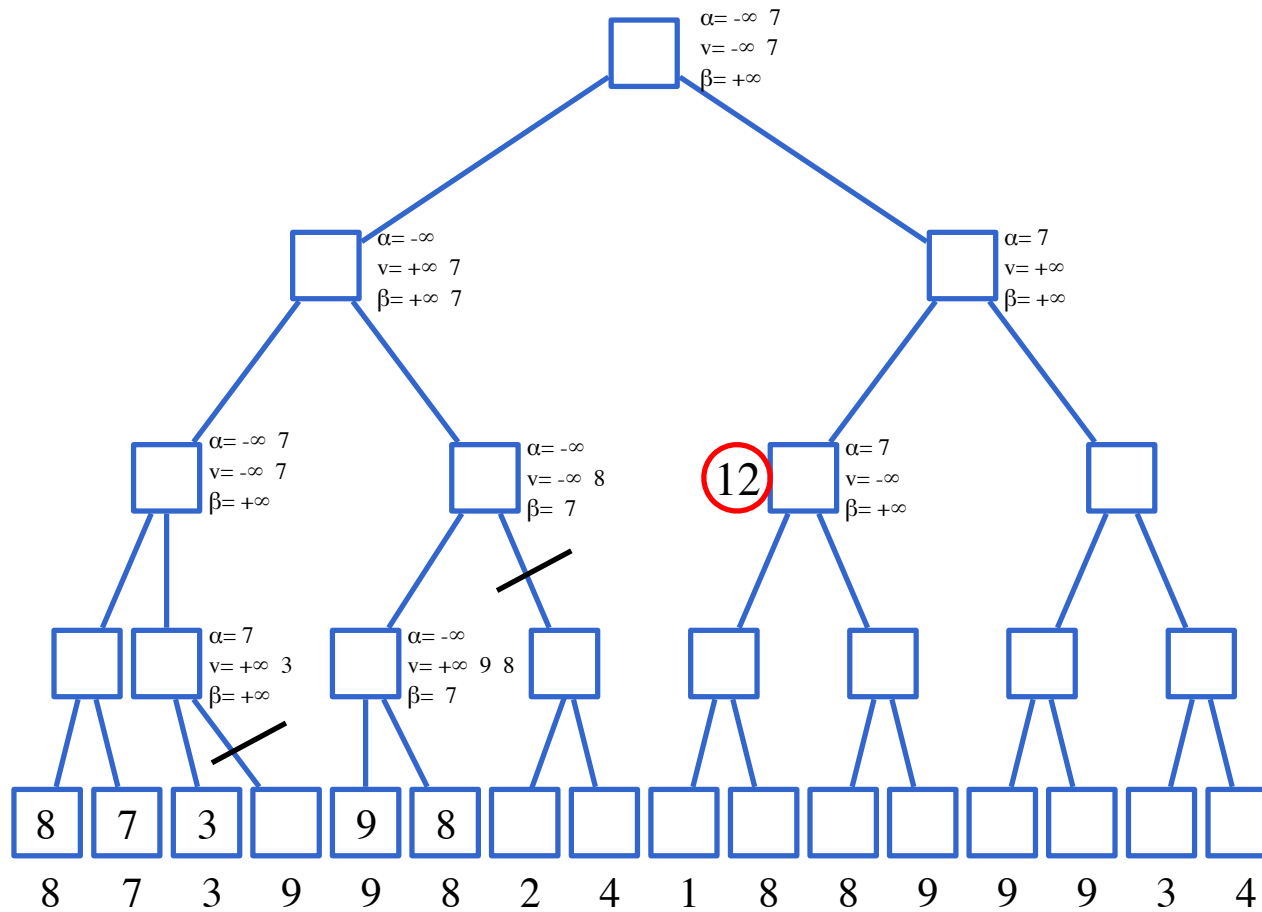


The diagram shows a minimax search tree. The root node is labeled '10' in a red circle. The tree has four levels of nodes. The leftmost branch is fully explored, with values propagated up: leaf nodes 8, 7, 3, 9; parent nodes 8, 7; grandparent nodes 8, 9; and the root node 10. Pruning is indicated by diagonal slashes on the branches leading to the rightmost leaf nodes of the first two levels on the left. The right half of the tree is shown but not fully explored, with values like $\alpha = -\infty$, $v = -\infty$, and $\beta = 7$ or 8 indicated for some nodes.

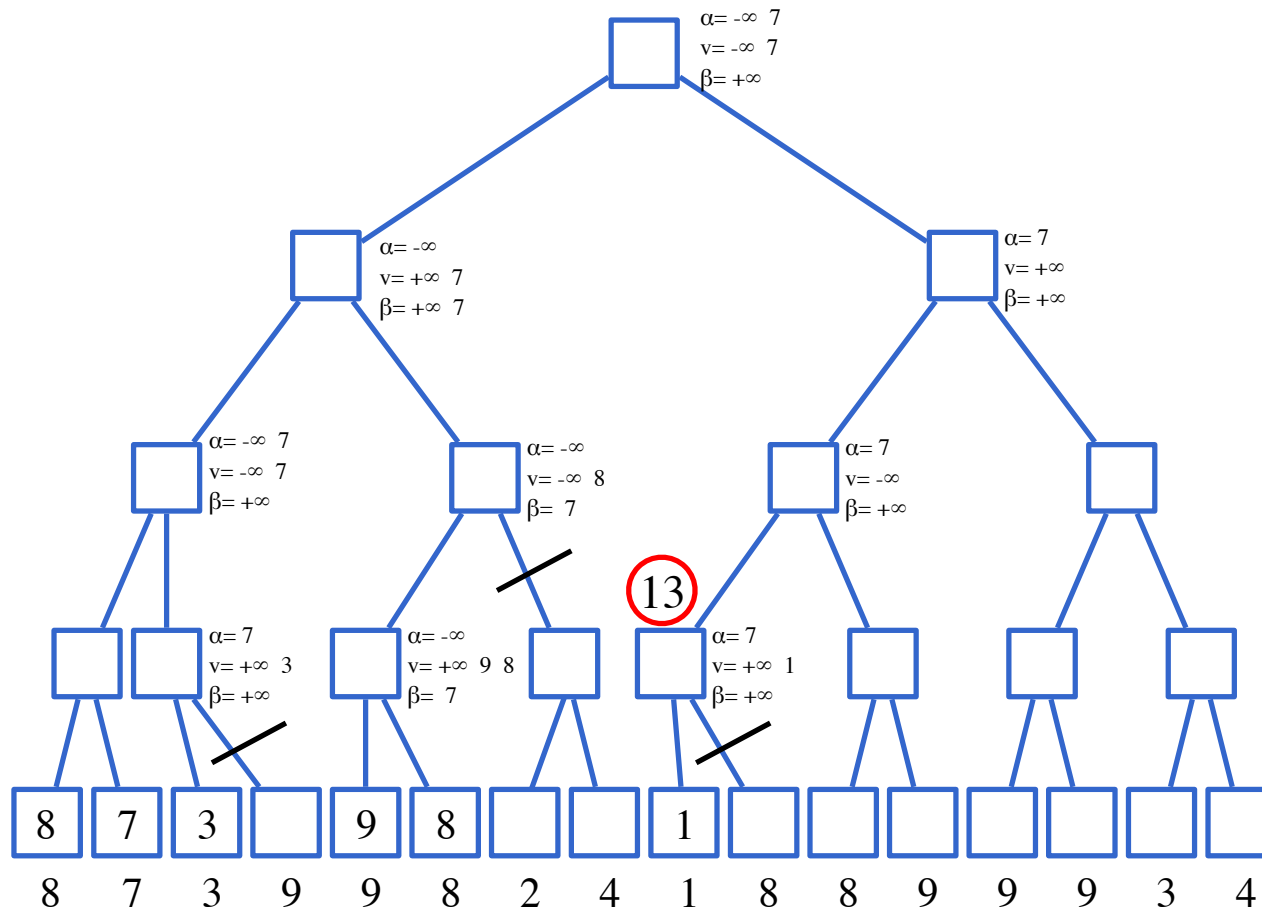
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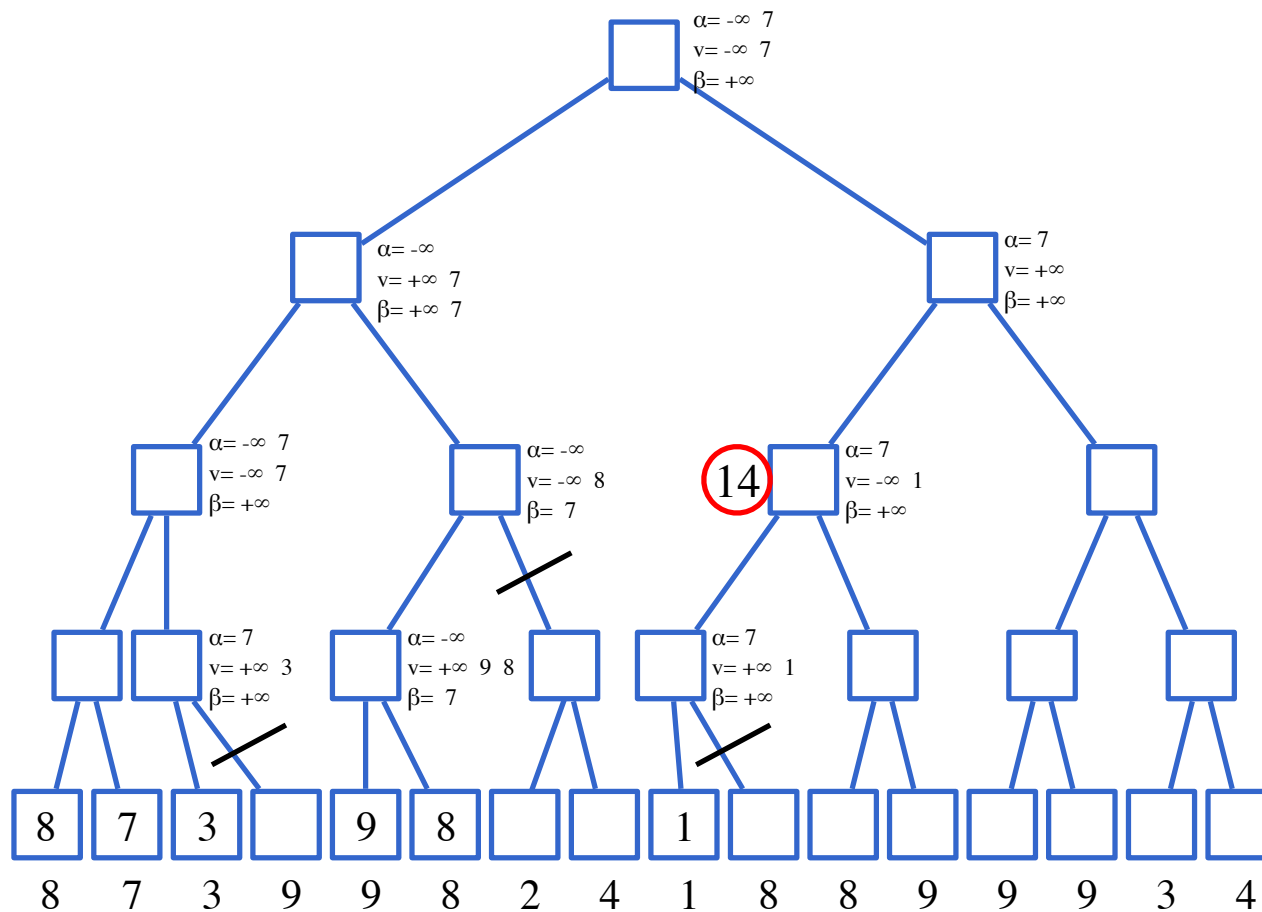
Exam 21/12/2011



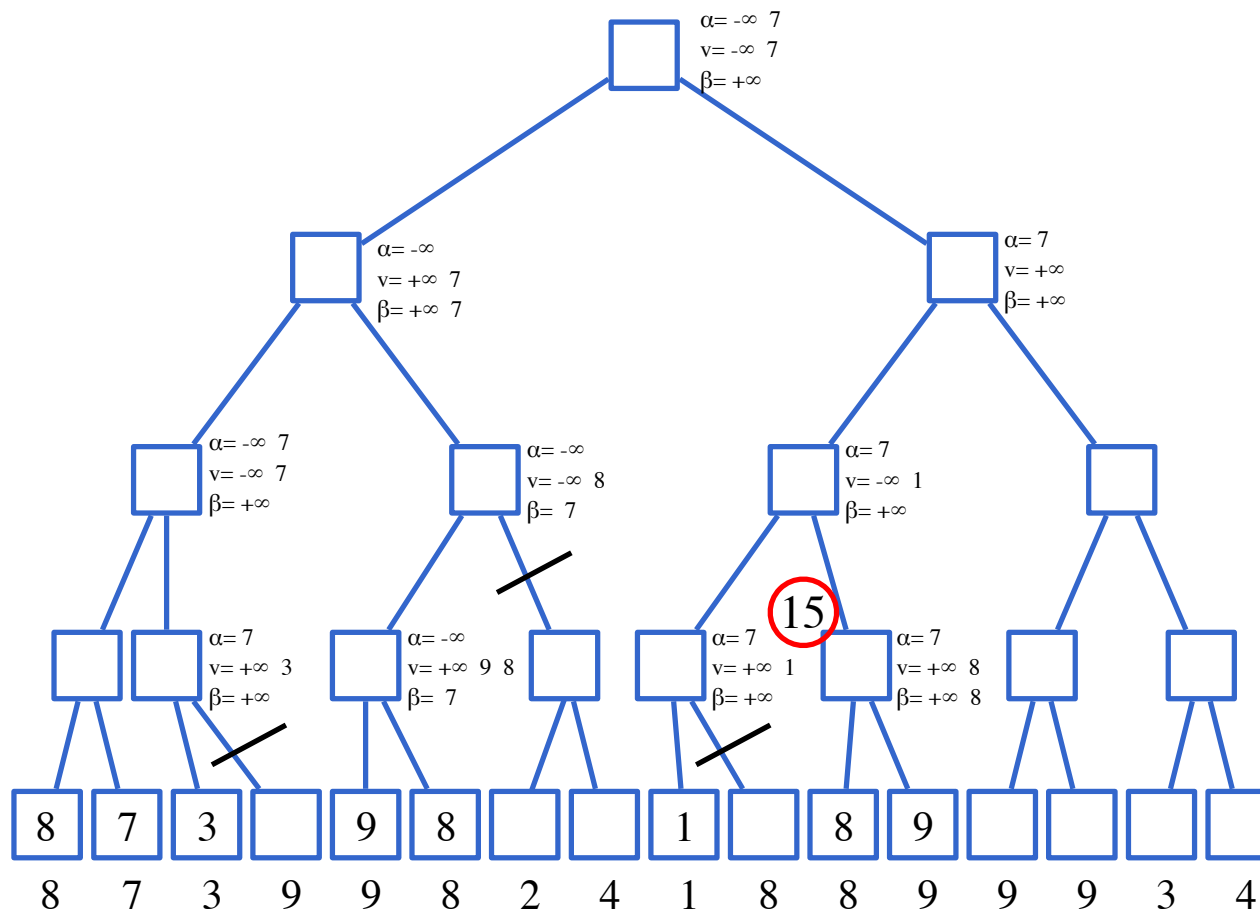
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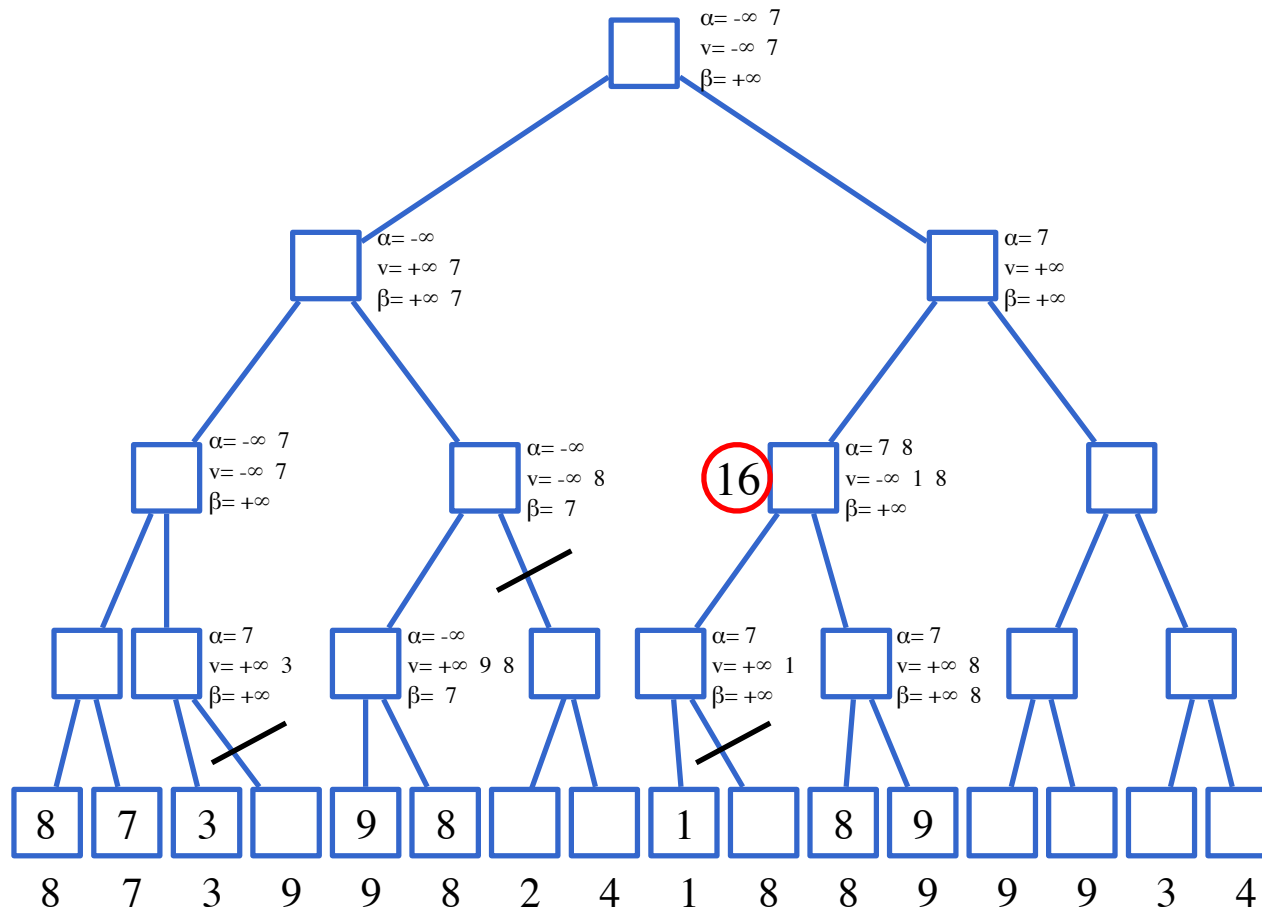
Exam 21/12/2011



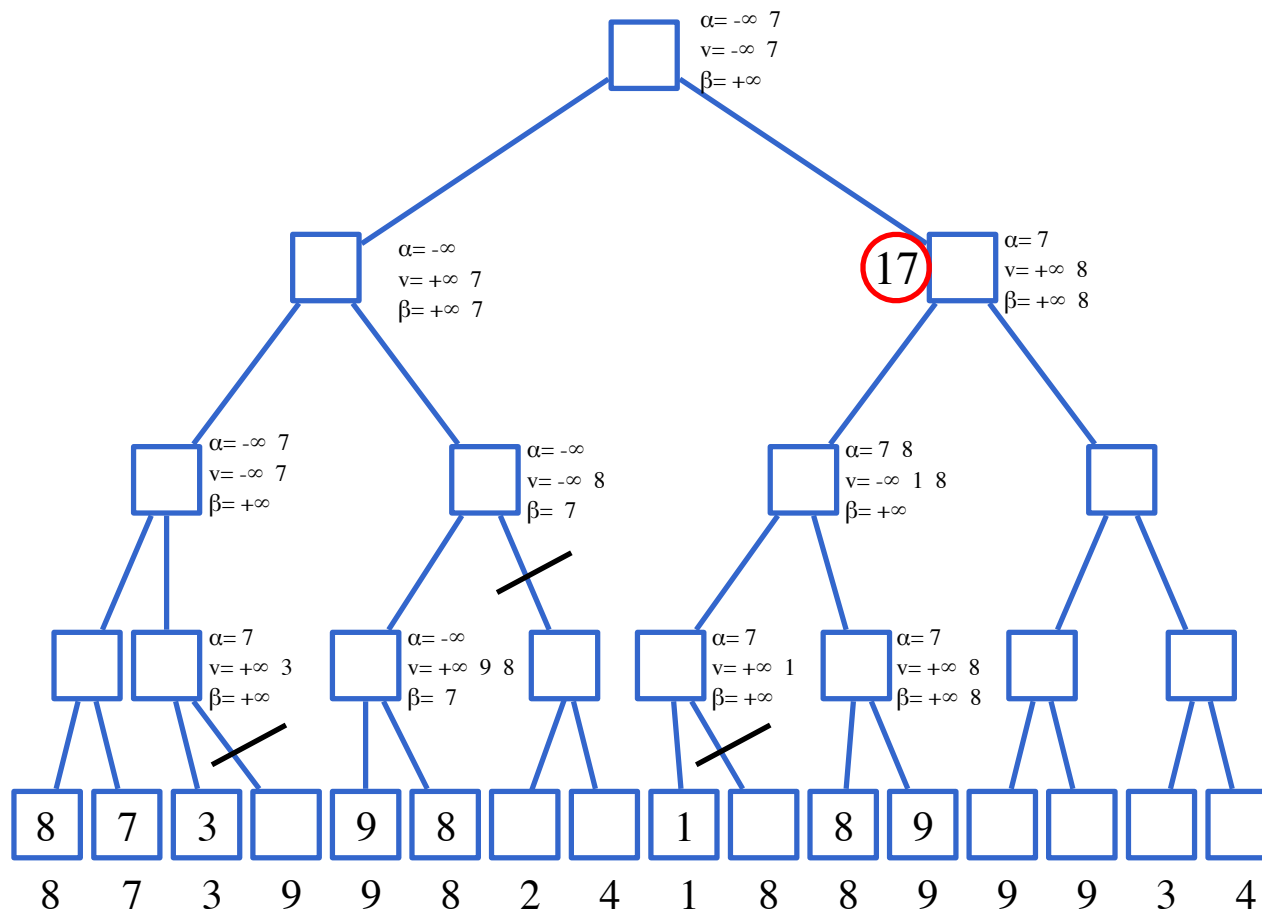
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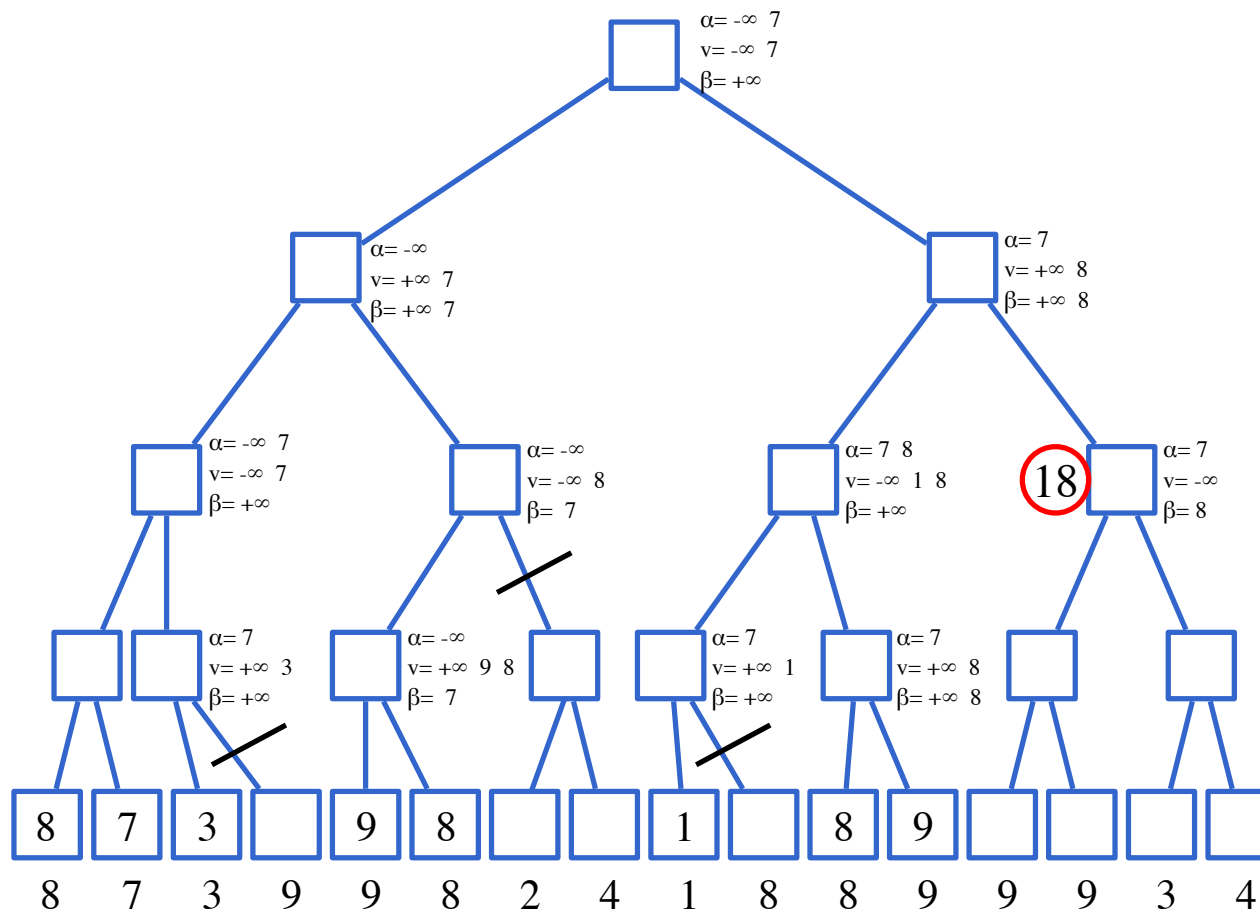
Exam 21/12/2011



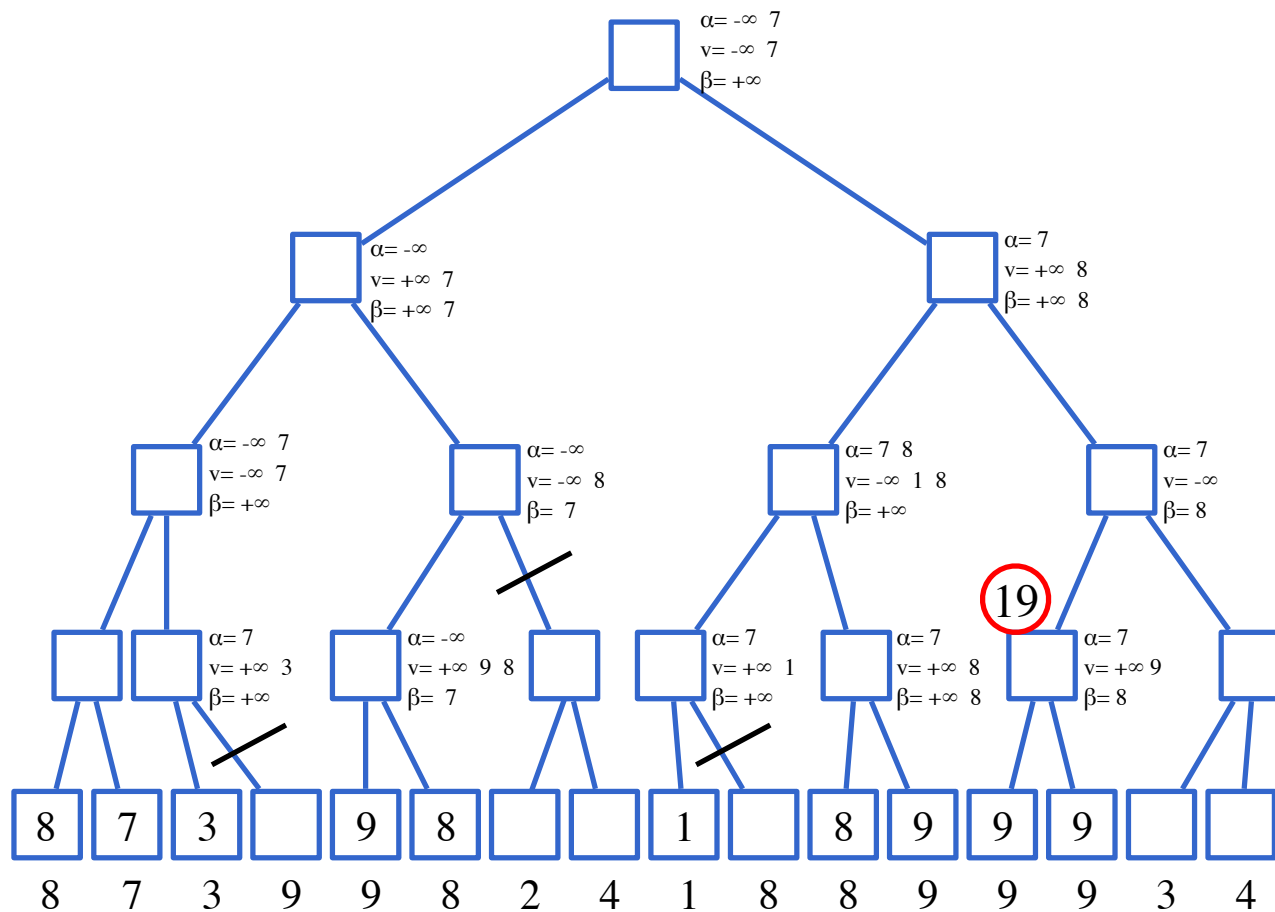
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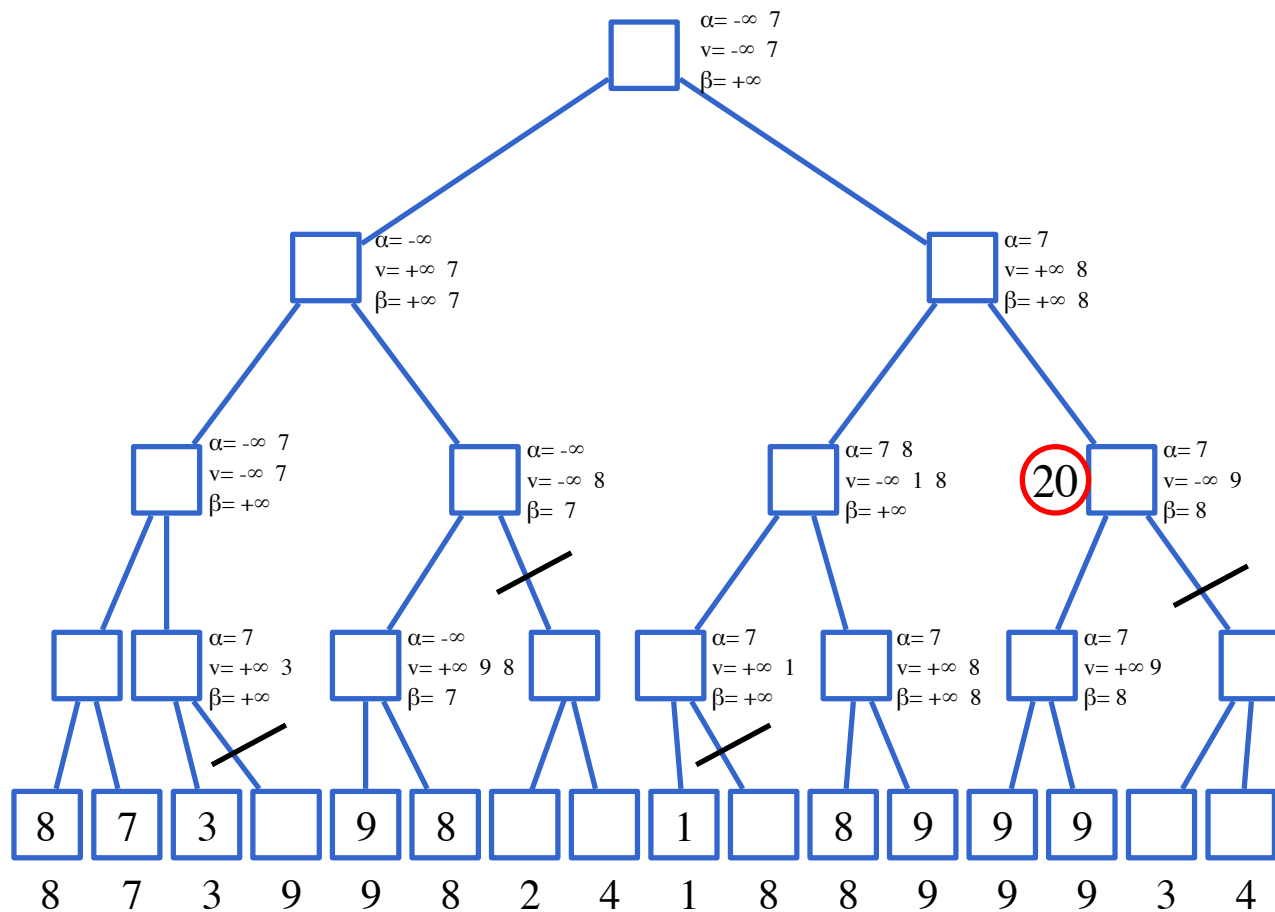
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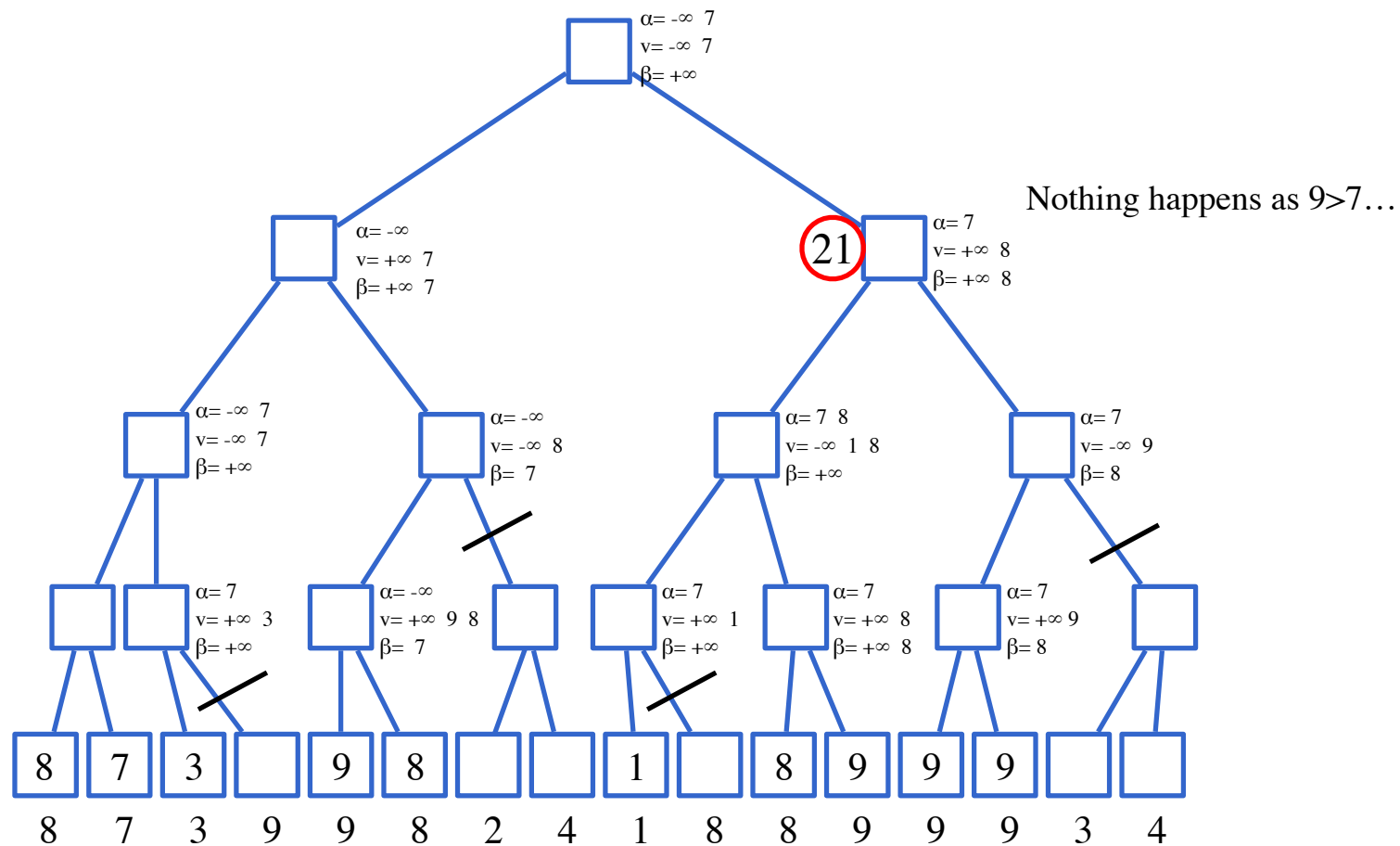
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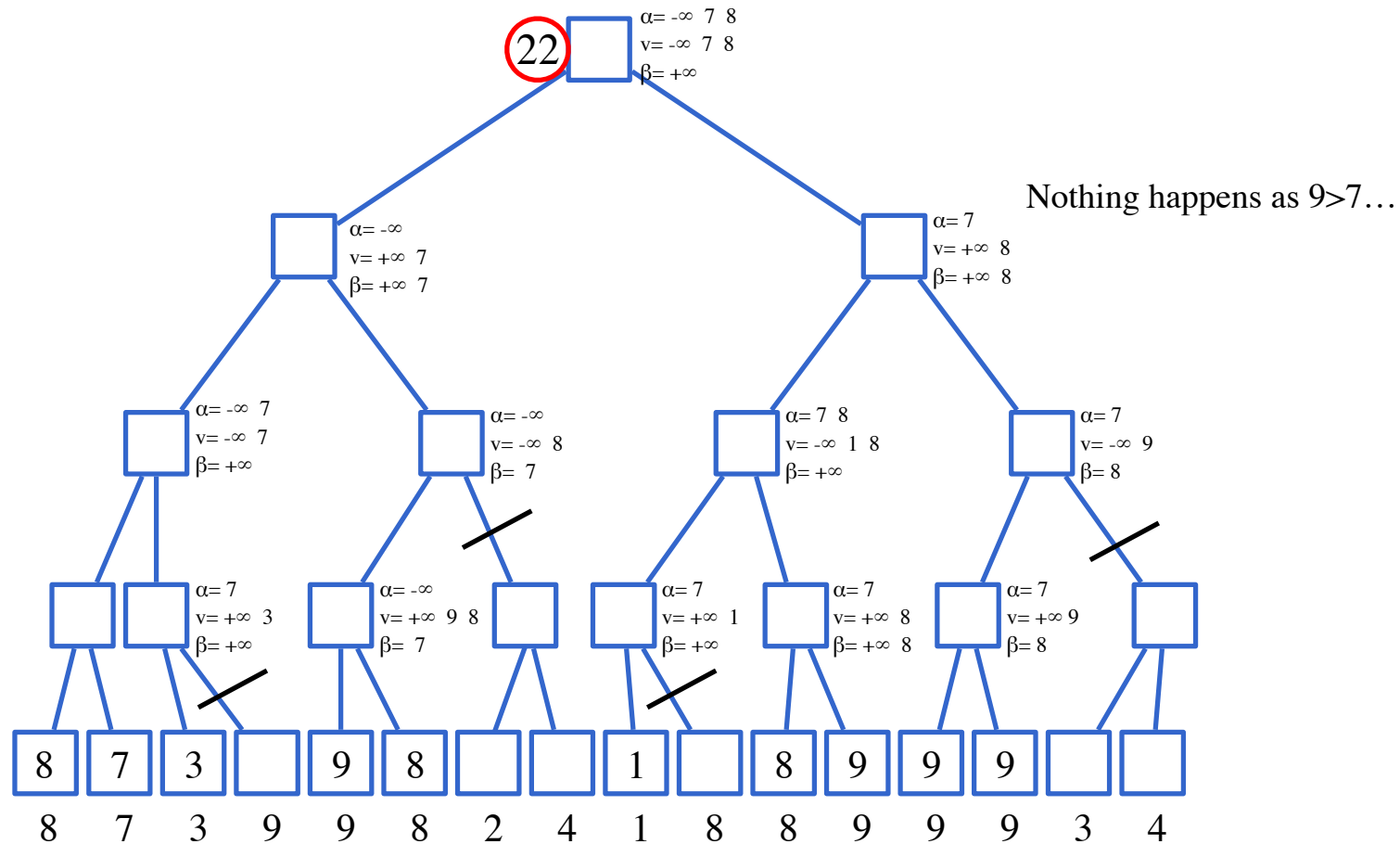
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Exam 21/12/2011



Exercise – min-max

Consider the following game:

- Two players have a single shared pile of coins
 - A pile can be divided only if the two resulting piles have a different number of coins.
 - The game ends when in each pile we have either one or two coins. At this point the player that cannot move loses.
-
- Build a MIN-MAX search tree in case MIN starts with a single pile of 7 coins.
 - Notation: each node of the search space is a list of numbers. Each number represents a pile and the number of coins in the corresponding pile

Exercise – min-max

- Min-Max:

